Process Characterization Report

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCR-007

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

AEX, anion exchange chromatography; ADCC, antibody-dependent cell-mediated cytotoxicity; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CI, confidence interval; Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; CV, coefficient of variation; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G isotype; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Cation exchange chromatography is the first of the two polishing steps in the A-Mab drug substance process. It uses a strong cation exchange resin in bind and elute mode and is the principal aggregate reduction step of the purification train. The step also clears host cell protein, residual DNA and leached Protein A. It forms no quality attribute of its own. Although some clearance of enveloped virus can be demonstrated across a cation exchange step, no viral clearance is claimed here. This report presents the characterization study executed against the paired protocol PCP-007 on a qualified scale-down model of the commercial 15,000 L process.

The pre-characterization risk assessment RA-001 carried 5 process parameters into the study, of which 4 were studied in a multivariate design and the remaining one was assessed one at a time. The screening study was a two-level full factorial in the four multivariate factors with 3 centre points, 19 runs in total. The response-surface study was a face-centred central composite design in the same four factors with 4 centre points, 28 runs in total. Pool aggregate, pool host cell protein and step yield were measured on every run.

Protein load produced the largest effect on all three responses. Over the characterized range of 10 to 40 g/L resin it raised pool aggregate by 0.60 % HMW, raised pool host cell protein by 116 ng/mg and reduced step yield by 5.2 percentage points. Load and wash conductivity acted on host cell protein in opposite directions and interacted ($p = 0.012$). Elution buffer pH and elution stop collect acted on aggregate and had no significant effect on host cell protein. Wash conductivity had no significant effect on aggregate.

The response-surface models for aggregate and host cell protein are adequate and predictive. Their predicted R^2 values are 0.82 and 0.83, and neither shows significant lack of fit against the centre-point pure error. The yield model identifies the protein load effect but its predicted R^2 of 0.20 does not support prediction, so yield is reported as an effect and is not used to set a range.

Over the characterized region, 68.7% of the modelled parameter combinations meet the in-process limits for both governed responses at the same time. Pool aggregate is the binding attribute. The proven acceptable ranges show the same ordering. With the other three parameters at their set-points, protein load is acceptable for aggregate from 10 to 36 g/L resin. When the other three parameters vary across their normal operating ranges the upper bound falls to 27.6 g/L resin, which is inside the upper edge of the load normal operating range of 32 g/L resin. Movement to the top of the load range therefore depends on where the other three parameters sit, and the multivariate design space rather than the individual ranges is the operating claim.

At commercial scale the four quality attributes this step governs meet their drug-substance criteria with wide margin. The tightest capability among them is host cell protein at a one-sided Cpk of 6.14, against 16.1 for aggregate. These estimates come from 2,000 simulated batches with every parameter inside its normal operating range, and they are confirmed at commercial scale during Stage 2 and not before.

At this step, 4 parameters were classified as well-controlled critical process parameters and 1 as a general process parameter. No parameter at this step was classified as a critical process parameter, because all four quality-linked parameters are held by instrumented control well inside the region the study characterized. During execution, 2 deviations were recorded. Both were retained after investigation and neither invalidated a design point. The outcomes of this step roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified by a platform process (CMC Biotech Working Group 2009). Its primary mechanism of action is antibody-dependent cell-mediated cytotoxicity, so the glycan attributes formed in the production bioreactor are linked to efficacy. The drug substance process comprises Steps 3 to 10, and the purification train runs Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and ultrafiltration and diafiltration. Commercial manufacture is at 15,000 L, and the process is being transferred under PTP-001 from the development site to the commercial drug substance facility.

Cation exchange chromatography is Step 7 of that train. Below its isoelectric point the antibody carries a net positive charge and binds the sulfonate ligand of the resin, and it is eluted by a step change in buffer pH that weakens the electrostatic attraction. Aggregate presents more positive surface charge to the ligand than monomer does, so it binds more strongly and elutes later. The step separates the two by where the eluate pool is stopped on the descending edge of the elution peak. Host cell protein and residual DNA are cleared in the load flow-through and the wash, and leached Protein A, which is acidic under the load conditions, passes through in the load.

In the nominal batch the step received a pool at 1.91 % HMW from low-pH viral inactivation and delivered 0.73 % HMW, a reduction of 2.6-fold. Over the same batch it reduced host cell protein from 9,125 ng/mg in the capture pool to 117 ng/mg, a reduction of 78-fold, at a step yield of 95.2%. No product quality attribute is formed at this step, and the charge and glycosylation variants are not separated by the step under platform conditions.

1.2 Objectives

The objectives of this study were the following.

(i) To quantify the relationship between each studied process parameter and pool aggregate, pool host cell protein and step yield over ranges wider than routine operation.

(ii) To define the multivariate region of the four studied parameters over which the attributes this step governs remain within their in-process limits.

(iii) To establish a proven acceptable range for each parameter against each governed attribute, both at the step set-point and with the remaining parameters varying across their normal operating ranges.

(iv) To classify every parameter of the step against the criteria in SOP-4001.

(v) To justify the ranges and the in-process controls that this step contributes to the control strategy for Stage 2.

1.3 Regulatory and scientific basis

The study was executed as Stage 1 process design under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8, the quality risk management principles of ICH Q9 and the development and manufacture principles of ICH Q11 (International Council for Harmonisation 2009, 2023b, 2012). Criticality is treated as a continuum rather than as a binary classification, so the quality attributes in scope carry a level of criticality and the attributes at high and very high criticality are called critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009).

Viral safety for A-Mab is claimed modularly across low-pH viral inactivation, anion exchange chromatography and small virus retentive filtration, and the cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a). Cation exchange chromatography contributes to that margin but is not credited in the claim, so no viral clearance study was performed on this step and none is reported here.

The process model, the quality attribute framework and the risk methodology follow the published A-Mab case study (CMC Biotech Working Group 2009). The paired protocol for this study is PCP-007, the risk basis is RA-001, and the campaign-level plan under which it was run is PCMP-001. The documents that frame this report and the document it rolls up into are listed in Table 1.1.

Table 1.1: Documents related to this report and the relationship each bears to it. {#tbl-related}

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-007	Process Characterization Plan (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Cation exchange polishing has been used to manufacture three previous licensed antibodies on this platform, X-Mab, Y-Mab and Z-Mab, and the resin, the buffer system and the mode of operation are unchanged for A-Mab. The large body of experience from those products established that the step reduces aggregate by a factor of two to three and host cell protein by one to two orders of magnitude when it is operated within platform conditions. This prior knowledge applies to A-Mab because the separation rests on the net charge of the antibody at the operating pH, and A-Mab has the isoelectric point and the charge distribution of the platform molecule. Characterization therefore had to bound the platform behaviour rather than discover it.

Prior knowledge also set the expectation for each factor before the study was run. Protein load determines what fraction of the ligand is occupied and how broad the elution peaks are. Near the capacity of the resin the monomer and aggregate peaks broaden and overlap, and weakly bound host cell protein finds fewer free sites from which the wash would otherwise have displaced it, so protein load was expected to act on aggregate clearance, on host cell protein clearance and on yield through one physical quantity. Load and wash conductivity set the ionic strength that shields the charge on the ligand and on the bound species, so a higher conductivity was expected to strip weakly bound host cell protein during the wash while the antibody stayed bound. Elution buffer pH sets the net charge on the antibody at the moment of elution, so a higher pH within the range was expected to bring monomer and aggregate off earlier and closer together and to reduce the resolution on which aggregate clearance depends. Elution stop collect decides how much of the descending edge enters the pool, and because the aggregate elutes on that edge it was expected to be the direct handle on the trade between aggregate and yield. Elution flow rate sets the residence time during elution, and at the particle size of this resin its effect on resolution was expected to be small.

Two attributes were expected to be insensitive to the operating parameters. Residual DNA is strongly negatively charged and is repelled by the cation exchange ligand, so it leaves in the load flow-through and its clearance is large and flat across the studied region. Leached Protein A and its fragments are acidic under the load conditions and do not bind the resin, and the fraction that is complexed with the antibody is released and passed through when the complex binds the ligand. The clearance of both is a property of the chemistry of the step rather than of any studied parameter, so neither was carried into the multivariate design. The step is credited with 1.5 log reduction of residual DNA and 1 log reduction of leached Protein A in the platform process description.

2.2 Quality attributes in scope

This step sets no quality attribute. The attributes in scope are the four that it clears, and each of them is formed upstream. They are listed in Table 2.1 with the acceptance criterion of the drug substance, the assigned level of criticality and the Tool #1 score from which that level was derived. Tool #1 scores the impact of an attribute on safety and efficacy against the uncertainty in that impact, and an attribute at high or very high criticality is called critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes cleared by the cation exchange step, with the drug-substance acceptance criterion and the assigned criticality. {#tbl-cqa}

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Aggregate is the attribute that governs this step. It carries the highest criticality of the four and the highest Tool #1 score, and cation exchange is the last step in the train that reduces it. Host cell protein is the second. It is cleared by three chromatography steps in series, so its acceptable level at the cation exchange pool depends on what the anion exchange step downstream removes, and Section 7 states the criterion that follows from that dependence. Residual DNA and leached Protein A are cleared across the step but were not measured as design responses, for the reason given above.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the step before any characterization run was executed. The ranking combined the potential impact of the parameter on a quality attribute with its potential to interact with other parameters, and the resulting score decided the study type rather than a classification (CMC Biotech Working Group 2009). Parameters that scored high on both axes were assigned to the multivariate design. Parameters that scored high on impact but low on interaction were assigned to univariate assessment. Parameters whose ranges are set by equipment design or by the platform buffer specification were not studied.

Under that rule, 4 parameters were assigned to the multivariate design, protein load, load and wash conductivity, elution buffer pH and elution stop collect. Elution flow rate was assigned to univariate assessment because prior knowledge indicated a small effect on resolution over the range the equipment permits and no mechanism by which it would interact with the other four. Column temperature, bed height and buffer composition are controlled to the platform specification and are identity-controlled and quality-controlled rather than characterized. The ranges taken into the study were set at two to three times the routine control range so that the study would bound process robustness and would leave room for later movement within the design space.

3 Materials and methods

3.1 Scale-down model and its qualification

The study was executed on a laboratory-scale chromatography system operated as a model of the commercial column. The model was designed on the established principles of chromatographic scaling under SOP-1001. Bed height, linear velocity, protein load per litre of resin, buffer composition and the wash and elution volumes normalized to column volumes were held at the commercial values, and only column diameter and the absolute volumes were reduced. The resin was packed and released against the column efficiency criteria of SOP-2008, using plate count and peak asymmetry, and the resin lot was drawn from the same manufacturing family as the material qualified for commercial supply.

Qualification of the model compared the small-scale system with the at-scale process on the attributes that enter and leave the step. The comparison covered pool aggregate, pool host cell protein, eluate pool volume and step yield, together with the column efficiency measures above. No difference was found between scales that was significant against the run-to-run variation of the small-scale system, and the model was accepted as representative of the commercial column for the purposes of characterization. Qualification data are held in the model qualification record referenced by PCP-007 and are not reproduced here.

The replicated centre point carries the qualification forward into the analysis. Every design in this study includes centre-point runs at the step set-point, and the variation among them is the pure error against which lack of fit is judged (Section 5). Because the model reproduces the commercial step, that pure error is a reasonable estimate of the run-to-run variation the commercial column would show at the same settings.

Two bounds apply to the model and they are stated here because every claim in this report rests on them. The model reproduces the commercial column at its target settings and was not qualified at the edges of the characterized ranges. Resin life-cycle effects are controlled under SOP-2008 and were not varied in this study, so the ranges reported here apply to resin within its qualified cycle count.

3.2 Operation

The step was operated in bind and elute mode. The column was equilibrated in the load buffer, loaded with the low-pH viral inactivation pool that had been adjusted to the load conductivity, washed at the same conductivity, and eluted by a step change to the elution buffer. Collection of the eluate pool began at a fixed threshold on the ascending edge of the elution peak and ended at the stop collect point on the descending edge, expressed in optical density units relative to the peak maximum. The column was then stripped, sanitized and stored under SOP-2008. The whole sequence is described in SOP-2010 and was followed without change in every run of this study.

Four operating parameters were varied by the design and are listed with their set-points in Table 4.1. Elution flow rate was varied only in the univariate assessment. Column temperature, bed height, buffer composition and the pH of the load and wash buffers were held at the platform specification. These inputs are controlled by identity testing and buffer release rather than by characterization, and they were verified before each run rather than varied within it.

3.3 Analytical methods

Every response in this study was measured by a validated method. Table 3.1 lists the controlled procedures and method validation records that apply to the step.

Table 3.1: Controlled procedures and analytical method validation records for the cation exchange step. {#tbl-sop}

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

Pool aggregate was measured as the high molecular weight fraction by size-exclusion high-performance liquid chromatography under AMV-3011, and this is the primary assay of the step. Pool host cell protein was measured by enzyme-linked immunosorbent assay under AMV-3012, whose validated precision is 9.5 % relative with a limit of quantitation of 1 ng/mg. Step yield was calculated from protein concentration by ultraviolet absorbance at 280 nm and the measured pool volume. Residual DNA was measured by quantitative polymerase chain reaction under AMV-3014, and leached Protein A by enzyme-linked immunosorbent assay under AMV-3016, whose validated precision is 6.5 % relative. Method performance is summarized in Appendix D.

The precision of the host cell protein assay matters to the interpretation of this study. Host cell protein is measured on a polyclonal reagent against a population of analytes whose composition changes with the operating conditions, and the assay is the least precise of the four. Its contribution to the observed variation of the pool result is quantified against the centre-point reproducibility in Section 5.

3.4 Sampling plan

Each run was sampled at four points. The adjusted load was sampled before application to the column and assayed for protein concentration, aggregate and host cell protein. The load flow-through and the wash were collected as composites and assayed for protein concentration and host cell protein. The eluate pool was sampled after mixing and assayed for protein concentration, aggregate, host cell protein, residual DNA and leached Protein A. Step yield was calculated from the load and pool protein masses.

The screening design carried 3 centre-point runs and the response-surface design carried 4 at the step set-point. All centre-point runs were executed with the same resin lot, the same buffer preparations and the same analytical sequence as the surrounding factorial runs, so that their spread measures the reproducibility of the study and not a separate experiment.

3.5 Statistical methods

The factors were coded so that the set-point of each is zero and the edges of its characterization range are minus one and plus one. Coding puts the factors on a common scale and makes the coefficients comparable, and every effect and coefficient in this report is reported on the coded scale unless the text says otherwise.

The screening data were analysed by ordinary least squares on a model containing the four main effects and the six two-factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response across the full characterization range of that factor. The response-surface data were analysed on the full quadratic model in the same four factors, containing the main effects, the two-factor interactions and the four pure quadratic terms. Terms were judged significant at $\alpha = 0.05$, and no term was removed from either model before reporting.

Model adequacy was assessed on four measures. R^2 states how much of the observed variation the model accounts for. Adjusted R^2 penalizes the number of terms and is the more informative of the two for a near-saturated design. Predicted R^2 is computed from the predicted residual error sum of squares, which refits the model with each run omitted in turn, and it is the measure that speaks to prediction rather than to description. The residual sum of squares was partitioned into lack of fit and pure error, using the replicated centre points for the latter, and lack of fit was tested by the ratio of the two mean squares. A model was accepted as predictive when its predicted R^2 was substantial and its lack of fit was not significant. All fitting followed SOP-1002.

Process capability was estimated by Monte-Carlo simulation of 2,000 batches of the whole process, with every parameter sampled inside its normal operating range and analytical variation applied to each result. Capability is reported as a one-sided Cpk against the drug-substance criterion, because all four attributes this step governs are impurity or size-variant ceilings with no lower limit.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the five parameters of the step with the set-point, the normal operating range, the range studied and the study type. The classification column reports the outcome of the study and is derived in Section 9.

Table 4.1: Process parameters of the cation exchange step, with set-point, normal operating range, characterization range, final classification and study type. {#tbl-params}

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	25	18–32	10–40	WC-CPP	multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	multivariate
Elution stop collect	OD desc.	1	0.7–1.3	0.5–1.5	WC-CPP	multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	univariate

The characterized ranges are wider than the normal operating ranges by a factor of two to three on each parameter. Protein load was studied from 10 to 40 g/L resin against a normal operating range of 18 to 32 g/L resin, and the upper edge approaches the point at which the resin capacity begins to limit the separation. Load and wash conductivity was studied from 3 to 7 mS/cm against a normal operating range of 4 to 6 mS/cm. Elution buffer pH was studied from 5.8 to 6.2 against a normal operating range of 5.85 to 6.15, which is the range over which the platform buffer system can be prepared and controlled. Elution stop collect was studied from 0.5 to 1.5 optical density units on the descending edge against a normal operating range of 0.7 to 1.3. Studying ranges wider than routine operation bounds the response of the step to a deviation and supports later movement within the design space, and it is the reason the characterized region is called the knowledge space.

The four multivariate factors are coded A to D throughout the report and in the appendices. Table 4.2 gives the correspondence.

Table 4.2: Coded factor legend for the screening and response-surface designs. {#tbl-legend}

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

4.2 Screening design

The screening study was a two-level full factorial in the four multivariate factors, 16 factorial runs augmented with 3 centre points for a total of 19 runs. The runs were executed in randomized order on a single resin lot. A full factorial in four factors estimates all four main effects and all six two-factor interactions without aliasing, which a fractional design of the same size could not do. The purpose of the screening study is to identify which parameters carry an effect and in which direction, and the design cannot resolve curvature because it holds only two levels of each factor. The screening design matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same four factors. It comprised 16 factorial runs, 8 axial runs and 4 centre points, a total of 28 runs. The axial points sit on the faces of the coded cube rather than beyond them, so no run was executed outside the characterization range and no extrapolation is needed to interpret the design. Adding the axial points estimates the four pure quadratic terms and turns the descriptive screening model into a predictive one. All four screening factors were carried into the response-surface design because all four proved active on at least one response (Section 5). The response-surface design matrix and the measured responses are given in Appendix B.

The two designs answer different questions and are reported as such. The screening design identifies the effects and their directions. The response-surface model is the predictive model of this step, and every design space, proven acceptable range and prediction in this report is taken from it.

4.4 Univariate assessment

Elution flow rate was assessed one at a time across 100 to 300 cm/hr with the other four parameters held at their set-points. Flow rate acts on resolution through residence time during elution, and at the particle size of this resin the prior expectation was a small effect over the range the equipment permits. No effect on pool aggregate, pool host cell protein or step yield was identified that could be separated from the reproducibility of the centre point, and the parameter was classified accordingly in Section 9. Because flow rate carries no demonstrated effect, it was not carried into the multivariate design and it does not appear in the design space of Section 6.

5 Results

5.1 Centre-point performance and reproducibility

The replicated centre points measure how reproducibly the scale-down model runs the step at its set-point, and they supply the pure error against which model lack of fit is judged. Table 5.1 gives the mean, the standard deviation and the coefficient of variation of each response over the 4 centre-point runs of the response-surface design. Table 5.2 gives the same for the 3 centre points of the screening design.

Table 5.1: Centre-point reproducibility of the response-surface design. {#tbl-cv-rsm}

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	0.771	0.043	5.58
Pool HCP (ng/mg)	4	134	18.7	13.9
Step yield	4	0.918	0.0289	3.15

Table 5.2: Centre-point reproducibility of the screening design. {#tbl-cv-scr}

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	3	0.785	0.0804	10.2
Pool HCP (ng/mg)	3	155	27.2	17.6
Step yield	3	0.925	0.0131	1.42

At the set-point the step delivered a pool at 0.771 % HMW aggregate and 134 ng/mg host cell protein at a yield of 91.8%. Step yield was the most reproducible response, at 3.2 % coefficient of variation, and aggregate followed at 5.6 %. Pool host cell protein was the least precise response at 13.9 %, and it was also the least precise response in the screening design at 17.6 %.

Most of that variation is analytical. The validated precision of the host cell protein assay is 9.5 % relative, so the assay alone accounts for the larger part of the 13.9 % observed at the centre point, and the process contribution is the smaller part. Host cell protein results near the limit of quantitation are therefore attributed to the assay unless a factor effect explains them, and the effects reported below are all far larger than this spread.

The centre points of the two designs agree with each other on all three responses, which supports treating the screening and response-surface studies as one body of data on one scale-down model. The screening centre points are more variable than those of the response-surface design on aggregate and host cell protein. That is expected from 3 replicates against 4, and the two standard deviations are not distinguishable at this replication.

5.2 Screening: factor effects

Table 5.3 summarizes the fit of the screening model for each response.

Table 5.3: Screening model summary for each response. Effects are estimated on the coded factors from a model containing the main effects and the two-factor interactions. {#tbl-fit-scr}

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	19	0.973	0.939	0.869	28.5	3.57e-05	0.096
Pool HCP (ng/mg)	19	0.936	0.857	0.467	11.8	0.000937	35.4
Step yield	19	0.924	0.829	0.498	9.74	0.00181	0.0115

The screening models describe the data closely, with R^2 from 0.924 to 0.973. The design is near-saturated, because ten terms and an intercept are estimated from 19 runs, so R^2 alone overstates how much the model knows. The predicted R^2 is the informative measure, and it falls to 0.47 for host cell protein and 0.50 for step yield. The screening study is therefore read here for which factors are active and in which direction, and the quantitative model of the step is the response-surface model of Section 5.3.

Aggregate levels were affected by protein load, elution buffer pH and elution stop collect. Table 5.4 gives the six largest effects.

Table 5.4: The six largest screening effects on pool aggregate. An effect is the change in the response across the full characterization range of the coded factor. {#tbl-eff-agg}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*

Protein load carried the largest effect. Across 10 to 40 g/L resin it raised pool aggregate by 0.60 % HMW ($p < 0.001$). Elution buffer pH raised it by 0.34 % HMW and elution stop collect by 0.32 % HMW, both in the direction prior knowledge expected. All three interactions among these factors were significant, the largest being protein load with elution buffer pH at 0.20 % HMW ($p = 0.004$). Load and wash conductivity had no detectable effect on aggregate. Its estimated effect was -0.018 % HMW ($p = 0.710$), which is smaller than the centre-point standard deviation of 0.043 % HMW.

Host cell protein levels were affected by load and wash conductivity and by protein load, in opposite directions, and a significant interaction between the two was identified. Table 5.5 gives the six largest effects.

Table 5.5: The six largest screening effects on pool host cell protein. {#tbl-eff-hcp}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	

Raising load and wash conductivity across 3 to 7 mS/cm lowered pool host cell protein by 127 ng/mg ($p < 0.001$). Raising protein load across 10 to 40 g/L resin raised it by 116 ng/mg ($p < 0.001$). The interaction between the two was -57 ng/mg ($p = 0.012$), about half the size of either main effect. Neither elution buffer pH nor elution stop collect had a significant effect on host cell protein, which is what the separation of the two mechanisms predicts. The wash removes weakly bound host cell protein before the elution begins, and the elution conditions therefore act on the size variants and not on the impurity load.

Step yield was affected by protein load alone. Table 5.6 gives the six largest effects.

Table 5.6: The six largest screening effects on step yield. {#tbl-eff-yield}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.0518	-0.0259	0.00287	-9.04	1.79e-05	***
BC	-0.0121	-0.00607	0.00287	-2.12	0.0672	
B	-0.0111	-0.00553	0.00287	-1.93	0.0899	
AD	-0.00914	-0.00457	0.00287	-1.59	0.15	
D	0.00683	0.00342	0.00287	1.19	0.268	
C	0.00666	0.00333	0.00287	1.16	0.279	

Across 10 to 40 g/L resin, protein load reduced step yield by 5.2 percentage points ($p < 0.001$). No other term reached significance, and the largest of them, the interaction of load and wash conductivity with elution buffer pH, was smaller than the yield spread of the screening centre points. Yield at this step is therefore a function of how heavily the column is loaded and of nothing else that was studied.

5.3 Response-surface models

Table 5.7 summarizes the fit of the full quadratic model for each response.

Table 5.7: Response-surface model summary for each response, from the full quadratic model in the four multivariate factors. {#tbl-fit-rsm}

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	28	0.975	0.949	0.823	36.9	3.85e-08	0.0749
Pool HCP (ng/mg)	28	0.966	0.929	0.827	26.3	3.07e-07	16.8
Step yield	28	0.825	0.637	0.196	4.38	0.00573	0.0188

The aggregate and host cell protein models are adequate for prediction. Aggregate is described with R^2 0.975, adjusted R^2 0.949 and predicted R^2 0.823. Host cell protein is described with R^2 0.966, adjusted R^2 0.929 and predicted R^2 0.827. In both cases the predicted R^2 is close to the adjusted R^2, which means the models predict a withheld run about as well as they describe the runs they were fitted to.

The step yield model is a weaker instrument and is treated as one. Its R^2 is 0.825 and its adjusted R^2 is 0.637, but its predicted R^2 is 0.20. The reason is that only one term of the fifteen is significant and the response spread is small beside the residual, so removing a run moves the fit. The model confirms the protein load effect that screening found and is used for nothing else in this report. No proven acceptable range and no part of the design space is taken from it.

Table 5.8 gives the full quadratic model for pool aggregate.

Table 5.8: Full quadratic model coefficients for pool aggregate, on the coded factors. {#tbl-coef-agg}

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.751	0.0259	29	3.36e-13	***
A	0.28	0.0176	15.9	6.77e-10	***
B	0.0276	0.0176	1.57	0.142	
C	0.172	0.0176	9.75	2.41e-07	***
D	0.157	0.0176	8.88	7.05e-07	***
AB	0.00964	0.0187	0.515	0.615	
AC	0.096	0.0187	5.13	0.000193	***
AD	0.0851	0.0187	4.54	0.000551	***
BC	-0.00211	0.0187	-0.113	0.912	
BD	0.0137	0.0187	0.733	0.477	
CD	0.0694	0.0187	3.71	0.00262	**
A^2	0.0798	0.0466	1.71	0.111	
B^2	-0.00875	0.0466	-0.188	0.854	
C^2	0.0264	0.0466	0.566	0.581	
D^2	0.0677	0.0466	1.45	0.17	

Term	Coef.	Std. err.	t	p-value	Sig.

Protein load remains the largest term, at a coded coefficient of 0.280 % HMW ($p < 0.001$), followed by elution buffer pH at 0.172 and elution stop collect at 0.157. The three interactions identified in screening are confirmed, and the interaction of protein load with elution buffer pH is again the largest at 0.096 ($p < 0.001$). None of the four quadratic terms is significant. The largest of them is the quadratic in protein load at 0.080 ($p = 0.111$). The aggregate surface over this region is therefore close to a plane with twist and shows no maximum, no minimum and no edge of failure inside the ranges studied.

Table 5.9 partitions the residual variation of that model.

Table 5.9: Analysis of variance for the pool aggregate model, with the residual partitioned into lack of fit and centre-point pure error. {#tbl-anova-agg}

Source	Sum sq.	df	Mean sq.	F	p-value
Model	2.89	14	0.207	36.9	3.85e-08
Residual	0.0729	13	0.0056	nan	nan
Lack of fit	0.0673	10	0.00673	3.63	0.158
Pure error	0.00556	3	0.00185	nan	nan

The model term is highly significant ($p < 0.001$). Lack of fit is not significant against the centre-point pure error, at F 3.63 on 10 and 3 degrees of freedom ($p = 0.158$). The quadratic model is therefore an adequate description of the aggregate surface over the characterized region. The test rests on 3 degrees of freedom for pure error, so it would detect only a substantial systematic departure.

Table 5.10 gives the full quadratic model for pool host cell protein.

Table 5.10: Full quadratic model coefficients for pool host cell protein, on the coded factors. {#tbl-coef-hcp}

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	141	5.81	24.3	3.21e-12	***
A	52.5	3.96	13.3	6.21e-09	***
B	-49.7	3.96	-12.6	1.21e-08	***
C	4.14	3.96	1.05	0.315	
D	-2.13	3.96	-0.538	0.6	
AB	-16.3	4.2	-3.87	0.00194	**
AC	-4.27	4.2	-1.02	0.328	
AD	-4.33	4.2	-1.03	0.321	
BC	-14	4.2	-3.33	0.00542	**
BD	0.547	4.2	0.13	0.898	
CD	-1.6	4.2	-0.381	0.71	

Term	Coef.	Std. err.	t	p-value	Sig.
A^2	-16.4	10.5	-1.56	0.142	
B^2	12.9	10.5	1.24	0.238	
C^2	2.31	10.5	0.221	0.829	
D^2	10.1	10.5	0.968	0.351	

Protein load and load and wash conductivity are the two large terms, at 52.5 ng/mg and -49.7 ng/mg respectively, and both are highly significant. Their interaction is confirmed at -16.3 ng/mg (p = 0.002). The model also resolves a second interaction, between load and wash conductivity and elution buffer pH, at -14.0 ng/mg (p = 0.005). That term was not distinguished from noise in the screening study, where its estimate carried the opposite sign at p = 0.330, and elution buffer pH has no main effect on host cell protein in either study. No mechanism is claimed for it. Its coefficient is about a quarter of either main effect, and it does not change the direction in which any parameter acts.

Table 5.11 partitions the residual variation of the host cell protein model.

Table 5.11: Analysis of variance for the pool host cell protein model. {#tbl-anova-hcp}

Source	Sum sq.	df	Mean sq.	F	p-value
Model	104,028.0	14	7,430.6	26.3	3.07e-07
Residual	3,670.2	13	282.3	nan	nan
Lack of fit	2,626.7	10	262.7	0.755	0.679
Pure error	1,043.5	3	347.8	nan	nan

Lack of fit is far from significant, at F 0.76 (p = 0.679). The lack-of-fit mean square is smaller than the pure-error mean square, which says that the scatter of the design points about the fitted surface is no larger than the scatter of the replicated centre points about their own mean. For this response the assay, and not the model form, sets the limit of what can be resolved.

The fitted surfaces are shown in Figure 5.1 and Figure 5.2. Each panel plots one response over two factors with the other two held at the design centre, which for every factor of this step is also the set-point. Figure 5.1 plots protein load against elution buffer pH, the pair that carries the aggregate separation. Figure 5.2 plots protein load against load and wash conductivity, the pair that carries the impurity clearance.

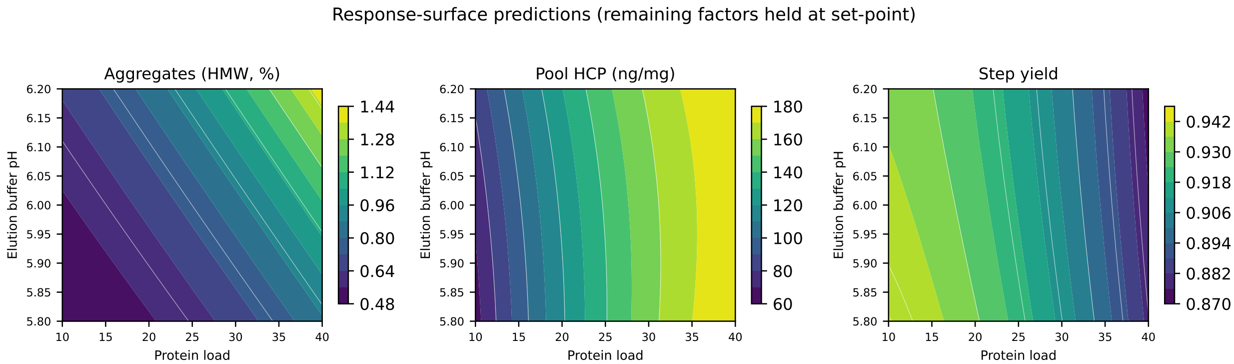

Figure 5.1: Response-surface predictions over protein load and elution buffer pH, with load and wash conductivity and elution stop collect held at the design centre.

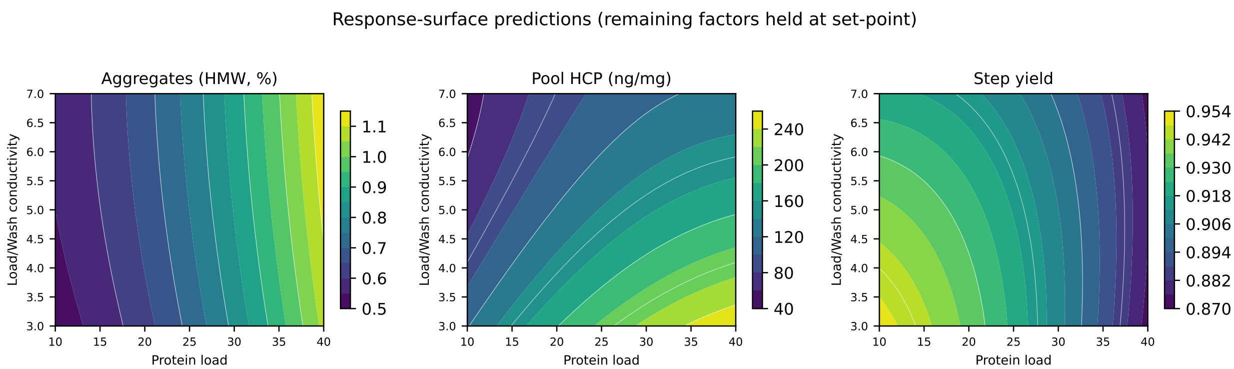

Figure 5.2: Response-surface predictions over protein load and load and wash conductivity, with elution buffer pH and elution stop collect held at the design centre.

In Figure 5.1 the aggregate contours run diagonally, which is the interaction between the two factors. Moving to the right along the protein load axis raises aggregate at every elution pH, and the rise is steeper at the top of the pH range than at the bottom. The host cell protein panel of the same figure is nearly flat in the vertical direction, because elution pH does not act on host cell protein. In Figure 5.2 the host cell protein contours run in the opposite sense on the two axes, and the aggregate panel of that figure is nearly flat in the vertical direction for the same reason in reverse.

Model diagnostics for the two predictive models are shown in Figure 5.3 and Figure 5.4. Each figure plots the standardized residuals against the predicted value, a normal quantile plot of those residuals, and the actual against the predicted value.

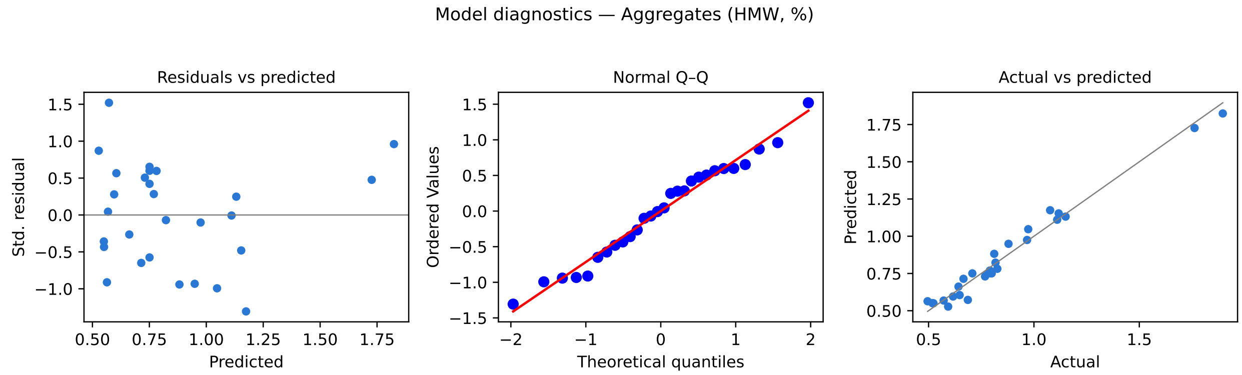

Figure 5.3: Diagnostics for the pool aggregate response-surface model.

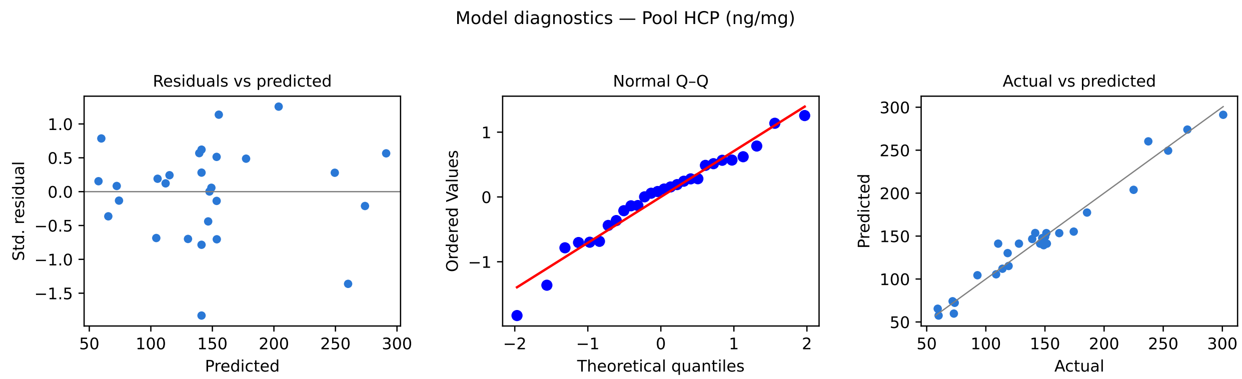

Figure 5.4: Diagnostics for the pool host cell protein response-surface model.

The residuals of both models are centred on zero and show no trend with the predicted value, so neither model is systematically wrong at one end of its range. The quantile plots are close to linear, which supports the normal-theory tests reported above. The actual against predicted panels lie along the identity line across the whole span of each response. No run was excluded from either fit, and no observation is influential enough to change a conclusion if it were removed.

5.4 Mechanistic interpretation

The three responses of this step move together because one physical quantity drives them, the fraction of the ligand that is occupied. Protein load sets that fraction. Near the capacity of the resin both the monomer and the aggregate peaks broaden, the aggregate front advances into the monomer peak, and weakly bound host cell protein finds fewer free sites from which the wash would have displaced it. This is why protein load is the largest term in all three models and why it acts to raise aggregate, to raise host cell protein and to lower yield at the same time. The step has one dominant handle, and moving it trades all three responses at once.

Elution buffer pH acts on the resolution of the separation rather than on its extent. A higher pH within the range lowers the net positive charge on the antibody, weakens its binding, and brings monomer and aggregate off the column earlier and closer together. Aggregate rises because the two populations are less well resolved when the pool is cut. The interaction with protein load follows from the same picture. When the column is heavily loaded the peaks are already broad, so a further compression of the separation costs more than it does on a lightly loaded column. This is the largest interaction in the aggregate model and it has the sign the mechanism predicts.

Elution stop collect is the direct handle on the trade between aggregate and yield, because the aggregate elutes on the descending edge of the peak. Stopping later takes more of that edge into the pool, which recovers monomer and admits aggregate. Its interaction with protein load and its interaction with elution buffer pH both act in the same direction as the main effect, and both have the same explanation. Anything that moves the aggregate forward into the tail makes the position of the stop point matter more.

Load and wash conductivity acts on the impurity clearance and not on the size variants, which the data confirm twice. It carries the largest single effect on host cell protein and no detectable effect on aggregate. The reason is that it operates during the load and the wash, before the elution begins. A higher ionic strength shields the charge on the ligand and strips weakly bound acidic species from it, while the antibody, which binds more strongly, stays bound. The interaction with protein load has the sign this predicts. A heavily loaded column carries more weakly bound host cell protein into the wash, so raising the wash conductivity removes more from it than it removes from a lightly loaded column.

None of the four quadratic terms is significant in any of the three models. Over the ranges studied the step behaves as a plane with twist rather than as a curved surface, so there is no optimum inside the region and no edge of failure at which a response turns. That result holds only over the characterized ranges. Approaching the capacity of the resin more closely than 40 g/L resin would be expected to introduce curvature, and this study does not bound the step there.

The clearance of residual DNA and of leached Protein A was confirmed across the step and was flat over the design. Both species are acidic or negatively charged at the load conditions and are excluded from the ligand by charge, so their removal depends on the chemistry of the resin and not on where within these ranges the step is operated. Neither attribute was carried into the multivariate model, and neither appears in the design space of Section 6.

6 Design space

The design space of this step is the region of the four well-controlled critical process parameters over which the two attributes the step governs remain within their in-process limits. It sits inside the characterized region, which is the knowledge space of Table 4.1, and it contains the normal operating ranges within which the commercial process is run. The knowledge space is what the study explored, the design space is the part of it that is known to give acceptable material, and the normal operating range is the smaller region the process is actually held in (CMC Biotech Working Group 2009; International Council for Harmonisation 2009).

Two criteria define the region. Pool aggregate must not exceed 1 % HMW and pool host cell protein must not exceed 193 ng/mg. Both are in-process limits rather than drug-substance criteria, and Section 7 derives them. Step yield carries no acceptance criterion and does not constrain the region, and residual DNA and leached Protein A are cleared with a margin that no combination of the studied parameters removes.

The size of the region was evaluated by predicting both responses from the response-surface models on an evenly spaced grid of 14,641 points across the coded cube. Of those points, 68.7% meet both criteria at the same time. Pool aggregate is the binding attribute, rejecting 21.1% of the region on its own, against 17.0% for host cell protein. The step therefore has a large but not unbounded design space, and the part of the knowledge space that is excluded lies at high protein load.

The two principal planes of the region are shown in Figure 5.1 and Figure 5.2. In the plane of protein load against elution buffer pH the boundary runs diagonally, because the two factors interact. The highest protein load that meets the aggregate limit falls as the elution pH rises, and at the low end of the pH range the whole load axis is acceptable. In the plane of protein load against load and wash conductivity the boundary runs the other way for host cell protein. A higher wash conductivity buys back protein load, because it removes more of the weakly bound host cell protein that a heavy load carries into the wash. Elution stop collect enters the region in the same sense as elution pH and is bounded by aggregate alone.

Three bounds apply to this design space and each of them limits what may be claimed from it. The region is defined on mean predictions from the fitted models, so at its boundary the probability of meeting the limit is about one half if the process and analytical variation are symmetric about the mean. Operating at the boundary is therefore not the same as operating inside the region, and the proven acceptable ranges of Section 7, which propagate the variation of the other parameters, are the practical statement of where the step may be run. The region is defined only over the ranges the study explored and says nothing about settings outside them. The region rests on a scale-down model qualified at the target settings of the commercial column and has not been confirmed at commercial scale at its edges.

Movement within the design space is not a change from the perspective of the regulatory filing, as stated in ICH Q8 (International Council for Harmonisation 2009). Movement is still a change from the perspective of the pharmaceutical quality system, and any planned move within the region is assessed prospectively under the change management system before it is made (International Council for Harmonisation 2008). Movement outside the region requires a new characterization study and a supplement.

7 Proven acceptable ranges

7.1 The acceptance basis

A proven acceptable range is the range of one parameter, with the others held or varied in a stated way, over which the response meets its acceptance criterion. The criterion this step is judged against is an in-process limit and not the drug-substance specification, and the reason is that the cation exchange pool is an intermediate. Its host cell protein content is reduced again by the anion exchange step, so comparing the pool against the drug-substance criterion would report a failure that does not exist.

The two limits were derived by carrying the drug-substance criterion back through the clearance the downstream train delivers and then dividing by an assurance factor. For host cell protein the drug-substance criterion is 100 ng/mg and the anion exchange step clears a further 6.2-fold, so the concentration above which the downstream train can no longer bring the batch to specification is 619 ng/mg at this pool. Dividing by an assurance factor of 3.2 gives the in-process limit of 193 ng/mg. For aggregate no step downstream of cation exchange reduces the high molecular weight fraction, so the ceiling at this pool is the drug-substance criterion itself, 5 % HMW. Dividing by an assurance factor of 5 gives the in-process limit of 1 % HMW. The assurance factor is what makes each of these an in-process control rather than a break-even point, because a limit set at the ceiling itself would leave the drug substance dependent on every downstream step delivering its nominal clearance exactly.

7.2 The two analyses

Each proven acceptable range was determined twice, and the two answers differ because they answer different questions. The first analysis moves one parameter across its characterization range with the other three held at their set-points. For this step the set-point of every multivariate parameter is also the centre of its characterization range, so this analysis reads directly off the fitted surface. The second analysis moves the same parameter across its range while the other three vary randomly within their normal operating ranges, and it reports the range over which the predictive bound still meets the criterion. The second is the range that applies to routine manufacture, because in routine manufacture the other parameters are not at their set-points.

Table 7.1 gives both analyses for every combination of governed attribute and multivariate parameter.

Table 7.1: Proven acceptable ranges for each governed attribute and each multivariate parameter, at the step set-point and with the other parameters varying within their normal operating ranges. {#tbl-par}

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Protein load	10–40	g/L resin	≤ 1	10–36	10–27.6
Aggregates (HMW, %)	Load/Wash conductivity	3–7	mS/cm	≤ 1	3–7	3–7
Aggregates (HMW, %)	Elution buffer pH	5.8–6.2	pH	≤ 1	5.8–6.2	5.8–6.05
Aggregates (HMW, %)	Elution stop collect	0.5–1.5	OD desc.	≤ 1	0.5–1.5	0.5–1.12
Pool HCP (ng/mg)	Protein load	10–40	g/L resin	≤ 193	10–40	10–28.4
Pool HCP (ng/mg)	Load/Wash conductivity	3–7	mS/cm	≤ 193	3.3–7	4.55–7
Pool HCP (ng/mg)	Elution buffer pH	5.8–6.2	pH	≤ 193	5.8–6.2	5.8–6.2
Pool HCP (ng/mg)	Elution stop collect	0.5–1.5	OD desc.	≤ 193	0.5–1.5	0.537–1.5

7.3 What the ranges show

Protein load is the parameter the ranges bind. Against pool aggregate it is acceptable from 10 to 36 g/L resin with the other three parameters at their set-points. When the other three vary across their normal operating ranges that upper bound falls to 27.6 g/L resin, which is below the upper edge of the load normal operating range of 32 g/L resin. The upper part of the routine load range is therefore not supported for aggregate when the other three parameters are simultaneously unfavourable.

The model states the size of that exposure directly. At protein load 32 g/L resin with elution buffer pH at 6.15 and elution stop collect at 1.3 optical density units, all three at the unfavourable edge of their normal operating ranges, the predicted pool aggregate is 1.25 % HMW against an in-process limit of 1 % HMW. At the set-point the prediction is 0.75 % HMW. The corresponding corner for host cell protein, protein load at 32 g/L resin with load and wash conductivity at 4 mS/cm, gives 194 ng/mg against an in-process limit of 193 ng/mg, which the prediction reaches but does not clear.

Three things bound what that finding means. The corner is the simultaneous worst case of three or two independent parameters and is not a condition the process is run at, which is why the normal operating ranges are treated as a box and the design space as the operating claim. The in-process limits themselves carry the assurance factors of 5 and 3.2 described above, so a prediction slightly above either limit is not a prediction of material outside the drug-substance specification. Over 2,000 simulated commercial batches with every parameter varying inside its normal operating range, the worst drug-substance aggregate result was 1.15 % HMW against a criterion of 5 % HMW, and the worst host cell protein result was 42.3 ng/mg against 100 ng/mg. The consequence for the control strategy is stated in Section 10. Protein load is held toward the lower part of its normal operating range when elution buffer pH or elution stop collect is at the top of its own.

The other three parameters are less constrained. Load and wash conductivity is acceptable across its whole characterization range for aggregate, which it does not affect. Against host cell protein it is acceptable from 4.55 to 7 mS/cm under normal operating range variation, so the lower part of its routine range depends on protein load being below the top of its own. Elution buffer pH and elution stop collect are acceptable across the whole of their characterization ranges at the set-point, and both narrow only at the top under normal operating range variation. None of the three is the parameter that binds the region.

7.4 Representative plots

Figure 7.1 shows the proven acceptable range for pool aggregate against protein load, which is its governing parameter. Figure 7.2 shows the same for pool host cell protein against load and wash conductivity. In each figure the left panel holds the other parameters at the set-point and the right panel varies them within their normal operating ranges and plots the median with a 95 % predictive band. The acceptance limit is the dashed line, the acceptable region of the parameter is shaded green, the set-point is the dotted vertical line and the normal operating range is the grey band.

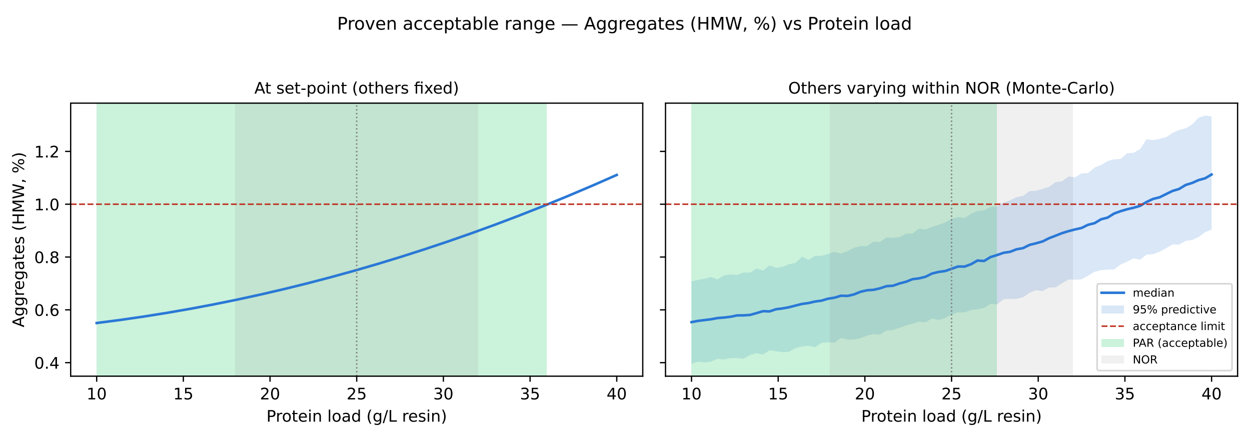

Figure 7.1: Proven acceptable range for pool aggregate against protein load, its governing parameter.

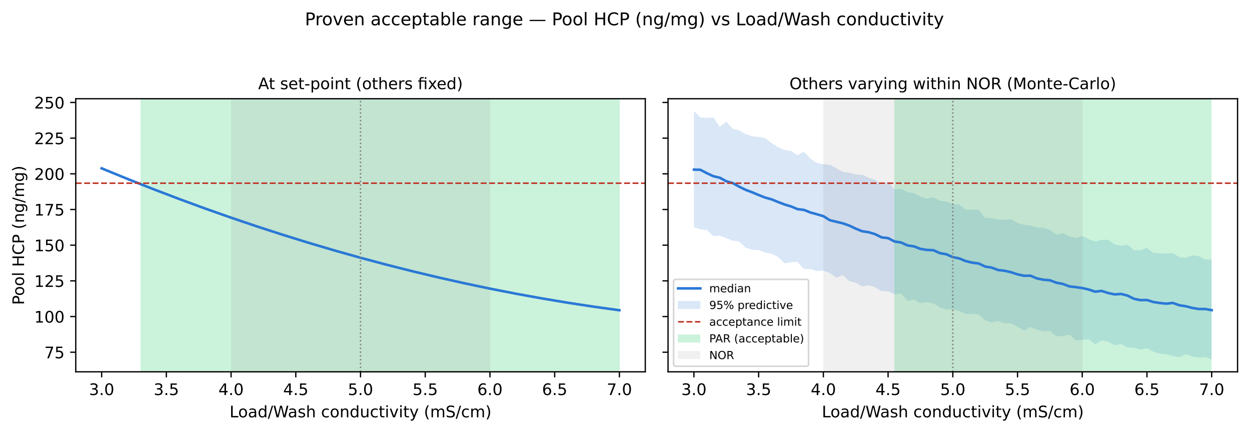

Figure 7.2: Proven acceptable range for pool host cell protein against load and wash conductivity, its governing parameter.

The two figures make the difference between the panels visible. In the left panel of Figure 7.1 the fitted line crosses the aggregate limit well above the normal operating range. In the right panel the predictive band is wide enough that its upper edge crosses the limit inside the normal operating range, and the green region ends there. That widening is the contribution of the other three parameters, and it is the reason the two proven acceptable range columns of Table 7.1 differ.

The univariate ranges of Table 7.1 and the multivariate region of Section 6 are consistent with each other and are used for different purposes. A univariate range answers what happens if one parameter moves, which is the question a deviation raises. The multivariate region answers where the process may be operated, which is the question the control strategy asks. Where the two disagree, as they do at the top of the protein load range, the multivariate region governs.

8 Process capability and robustness

Capability at commercial scale was estimated by simulating 2,000 batches of the whole process with every parameter sampled inside its normal operating range and analytical variation applied to each result. Table 8.1 gives the outcome for the four attributes this step governs.

Table 8.1: Commercial-scale capability for the quality attributes cleared by the cation exchange step, from the Monte-Carlo simulation of the whole process. {#tbl-cap}

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.1
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2,785.3

All four attributes meet their drug-substance criteria with substantial margin. Aggregate runs at 0.75 % HMW against a criterion of 5 % HMW, at a one-sided Cpk of 16.1. Host cell protein runs at 15.2 ng/mg against 100 ng/mg, at a Cpk of 6.14. Residual DNA and leached Protein A are at Cpk 10.8 and 2785 respectively, the second because the attribute is cleared to three orders of magnitude below its criterion.

Host cell protein is the tightest of the four and is the attribute to watch. Its capability of 6.14 is the smallest margin among the attributes this step governs, and its standard deviation of 4.6 ng/mg is the largest relative to its mean. Both follow from the same two facts, that host cell protein enters the step at 9,125 ng/mg and that the assay measuring it is the least precise of the four.

None of these four capabilities is the tightest in the drug substance. The smallest one-sided Cpk across every attribute of the process is 1.51, and it belongs to the cumulative viral clearance claim, which this step does not contribute to. The attributes this step governs are therefore not the constraint on the process, and PCMR-001 consolidates that comparison across the campaign.

Capability at this step is shared with other steps and is not delivered by cation exchange alone. Host cell protein is cleared by Protein A capture, by this step and by anion exchange, and the reports for those steps are PCR-005 and PCR-008. Residual DNA and leached Protein A are cleared by the same three steps. Aggregate is the exception. It is formed in the production bioreactor, raised slightly by the low-pH hold, and reduced only here, so the aggregate capability of the drug substance rests on this step in a way that the impurity capabilities do not. Aggregate is therefore the binding attribute of the design space, and its in-process limit carries the larger of the two assurance factors.

Two bounds apply to every number in this section. The estimates come from qualified scale-down models with every parameter inside its normal operating range, and they are not measurements of the commercial process. They are confirmed at commercial scale during Stage 2 process performance qualification and not before.

9 Parameter classification

Each parameter was classified against SOP-4001 from the effects demonstrated in Section 5 and from the risk that the parameter leaves the design space in routine operation. A parameter whose variation affects a critical quality attribute is a critical process parameter or a well-controlled critical process parameter, and the difference between the two is the risk of falling outside the design space rather than the size of the effect (CMC Biotech Working Group 2009). The classifications are given in Table 4.1 and the reason for each is below.

- Protein load was classified as a well-controlled critical process parameter. It carries the largest effect on both governed attributes and on step yield, and its proven acceptable range against aggregate is the one that binds the design space. It is set by the measured mass of the viral inactivation pool and the measured column volume, both of which are known before the load begins, so a load outside its range is prevented rather than detected. This is the parameter with the narrowest margin at this step.
- Load and wash conductivity was classified as a well-controlled critical process parameter. It carries the largest effect on pool host cell protein and no effect on aggregate. It is measured in line during the load and the wash and is controlled by buffer preparation and in-line dilution, and the measurement is continuous rather than at a single point, so an excursion is detected while the step is running.
- Elution buffer pH was classified as a well-controlled critical process parameter. It raises pool aggregate across its range and interacts with protein load in doing so. It is set by buffer preparation and confirmed by buffer release testing before use, and the characterization range is already the range over which the platform buffer system can be prepared, so the risk of operating outside it is low.
- Elution stop collect was classified as a well-controlled critical process parameter. It raises pool aggregate across its range and is the direct handle on the trade between aggregate and yield. It is set by the ultraviolet detector on the descending edge of the elution peak, which is an instrumented decision taken automatically at a fixed threshold, so it does not depend on operator judgement.
- Elution flow rate was classified as a general process parameter. The univariate assessment identified no effect on any measured response over 100 to 300 cm/hr that could be separated from the reproducibility of the centre point. It is controlled by the pump and is monitored, but it is not linked to a quality attribute and it does not enter the design space.

Of the 5 parameters at this step, 4 are therefore well-controlled critical process parameters and 1 is a general process parameter. No parameter was classified as a critical process parameter. All four quality-linked parameters are set or measured by instrumented systems, each is verified before or during the step rather than after it, and none of them depends on a manual adjustment during operation. The risk that any of them leaves the design space in routine manufacture is correspondingly low, which is the criterion that separates a well-controlled critical process parameter from a critical one.

10 Contribution to the control strategy

The control strategy for this step has four elements, the parameter ranges, the in-process controls on the pool, the life-cycle controls on the column, and the release testing that confirms the outcome at the drug substance. Each of them follows from a result in this report.

The parameter ranges are the normal operating ranges of Table 4.1, and they are the ranges the batch record specifies. They sit inside the design space of Section 6 with one qualification. Protein load at the top of its normal operating range is supported only when elution buffer pH and elution stop collect are not simultaneously at the top of theirs, and the batch record therefore carries the load target as a value calculated from the measured mass of the viral inactivation pool rather than as a range to be filled. Elution flow rate is controlled as a general process parameter and is monitored rather than restricted by this study.

Two in-process controls are placed on the eluate pool. Pool aggregate is tested by size-exclusion chromatography under AMV-3011 against an in-process limit of 1 % HMW. Pool host cell protein is tested by enzyme-linked immunosorbent assay under AMV-3012 against an in-process limit of 193 ng/mg. Both limits are derived in Section 7 and both are tighter than the drug-substance criterion carried back to this pool, so a pool that meets them leaves the downstream train the clearance it was credited with. Step yield is recorded as a process attribute and is monitored for consistency, and it carries no acceptance criterion because it is not linked to a quality attribute.

The life-cycle controls sit on the column rather than on a run. The resin is packed and released against column efficiency criteria and is used within a qualified cycle count under SOP-2008. Resin life-cycle was not varied in this study, so the ranges reported here are conditional on the column remaining inside that qualification.

One control follows from a deviation. DEV-007-01 arose because a buffer lot was used after its expiry date, and buffer expiry is now verified against the release record at preparation under SOP-2103 rather than at use. The investigation is in Section 13.

Release testing at the drug substance confirms the four attributes this step governs. The step does not act alone on any of them except aggregate. Host cell protein, residual DNA and leached Protein A are cleared by Protein A capture, by this step and by anion exchange, and the ranges and controls those steps contribute are in PCR-005 and PCR-008. The glycan and charge-variant attributes are formed in the production bioreactor and are not separated by this step, so they are controlled by PCR-003. The viral clearance claim rests on PCR-006, PCR-008 and PCR-009 and takes no credit from cation exchange. The elements listed here are consolidated with those of the other steps in PCMR-001, which is the document that states the control strategy for the process as a whole.

11 Discussion

The step is now characterized as a separation whose behaviour is governed by one physical quantity. Protein load sets the fraction of the ligand in use, and through it the breadth of the elution peaks, the resolution between monomer and aggregate, the number of free sites from which the wash can displace host cell protein, and the losses at both edges of the pool. Every large term in every model of Section 5 is an expression of that one quantity or of a factor that modifies it. The step therefore has a single dominant handle, which makes it easy to control and makes the consequence of moving that handle predictable in all three responses at once.

The second finding is that the two governed attributes are separated by mechanism, and this is what makes the design space tractable. Load and wash conductivity acts during the load and the wash and moves host cell protein without touching aggregate. Elution buffer pH and elution stop collect act during the elution and move aggregate without touching host cell protein. Only protein load acts on both. A design space in four parameters is therefore almost two design spaces in three, sharing one axis, which is why its boundary is simple enough to state in two planes.

Confidence in the scale-down model rests on three observations. The centre points of the two independent designs agree with each other on all three responses. The centre-point reproducibility is small beside every effect the study reports, so the design was able to resolve what it was built to resolve. Lack of fit is not significant for either predictive model against that centre-point pure error, which says the quadratic model form is adequate and not merely convenient. What the model has not shown is that the small-scale column behaves like the commercial column at the edges of the ranges, because it was qualified at the target settings.

During execution, 2 deviations were recorded and both were retained. Neither invalidated a design point, and neither required a study to be re-executed. The buffer expiry deviation touched an attribute that is measured in line during the run, and the temperature excursion touched a run whose result the model predicts within its own residual spread. The investigations are in Section 13 and the one control change that followed is in Section 10.

Four limitations bound what this report claims, and each is managed forward rather than resolved here. The scale-down model was qualified at the target settings of the commercial column and not at the edges of the characterized ranges, so the design space is a claim about the model at its edges and about the commercial column at its centre. Stage 2 confirms the centre at scale, and movement toward an edge is assessed under change control before it is made. The response-surface models predict mean levels, so the boundary of the design space carries about one half the reliability of its interior, and the proven acceptable ranges under normal operating range variation are the statement that accounts for this. Resin life-cycle was held constant and is controlled separately under SOP-2008, so the ranges reported here apply to a column inside its qualified cycle count. The step yield model has a predicted R^2 of 0.20 and is used only to confirm the protein load effect, so no yield claim in this report depends on prediction.

One result in this report is reported without an explanation. The response-surface model for host cell protein resolves an interaction between load and wash conductivity and elution buffer pH that the screening study did not find, and elution buffer pH has no main effect on host cell protein in either study. The term is about a quarter of the size of either main effect and it changes no direction and no range, so it is reported and carried forward rather than acted on. It would be resolved by replication at the affected corners if a future study revisits this step.

12 Conclusions

The cation exchange step was characterized over ranges two to three times wider than routine operation in 47 executed runs across two designs. Protein load, elution buffer pH and elution stop collect determine pool aggregate, load and wash conductivity and protein load determine pool host cell protein, and protein load alone determines step yield. The response-surface models for the two governed attributes are adequate and predictive, with predicted R^2 of 0.82 and 0.83 and no significant lack of fit.

A design space was defined over the four multivariate parameters, covering 68.7% of the characterized region, with pool aggregate as the binding attribute. Proven acceptable ranges were established for every combination of governed attribute and multivariate parameter, both at the set-point and under normal operating range variation, and they identify protein load as the parameter that binds the region. In total, 4 parameters were classified as well-controlled critical process parameters and 1 as a general process parameter, and none as a critical process parameter.

At commercial scale the four quality attributes this step governs meet their drug-substance criteria, the tightest among them being host cell protein at a one-sided Cpk of 6.14. These estimates rest on qualified scale-down models with every parameter inside its normal operating range and are confirmed at commercial scale during Stage 2. In the course of the study, 2 deviations were recorded and both were retained after investigation. The step is ready to enter process performance qualification within the ranges stated here, and its outcomes roll up into PCMR-001.

13 Deviations from the plan

During execution of this study, 2 deviations from PCP-007 were recorded. Both were detected after the affected run rather than during it, both were investigated to a root cause, and both were dispositioned as retained. Table 13.1 is the register.

Table 13.1: Deviations recorded during execution of the cation exchange characterization study. {#tbl-dev}

Deviation	Summary	Detected during	Disposition
DEV-007-01	Expired Tris load/wash buffer lot used in screening run 6	post-execution documentation review	retained
DEV-007-02	Column inlet temperature excursion during RSM run 14	equipment history review	retained

13.1 DEV-007-01 — expired load and wash buffer lot

The Tris load and wash buffer lot LOT-BUF-2287 was used to prepare screening run 6 after its expiry date of 2026-02-11. The deviation was found during the post-execution documentation review and not at the time of use. The root cause was that the expiry date of the lot was not verified when the buffer was prepared, and the preparation record carried no check against the release record.

The retained sample of the lot was re-tested under SOP-2103, which authorises retained-sample re-test on an expiry deviation. The re-test returned a pH of 5.02 and a conductivity of 5.06 mS/cm for the buffer as prepared, so the lot met its own attributes after the expiry date had passed.

Screening run 6 was the factorial corner at protein load 10 g/L resin, load and wash conductivity 7 mS/cm, elution buffer pH 5.8 and elution stop collect 1.5 optical density units. The conductivity a run is operated at is set by adjusting the base buffer to the design target and is measured in line throughout the load and the wash, so the attribute the design varies was verified during the run independently of the lot record. The run is also not anomalous against the model fitted to the whole design. Its standardized residual is 0.68 in the screening aggregate model, against a largest absolute standardized residual of 1.99 across the 19 runs, and 0.11 in the screening host cell protein model against a largest absolute value of 1.32.

The deviation was dispositioned as retained. Run 6 was kept in the analysis and no design point was invalidated. The control change that followed the investigation is stated in Section 10.

13.2 DEV-007-02 — column inlet temperature excursion

The column inlet temperature of the temperature-controlled chromatography chamber EQ-CHR-118 reached 26.4 °C and stayed above the chamber set-point for 38 minutes during response-surface run 14. The excursion was found during the equipment history review after the design had been executed. The root cause was a drift of the chamber controller set-point. The chamber was inside its calibration interval when the run was executed, with calibration next due on 2026-05-30, so the temperature record itself is reliable.

Temperature acts on the strength of the electrostatic interaction between the antibody and the ligand, and a warmer column would be expected to elute the antibody slightly earlier and to compress the separation between monomer and aggregate. Run 14 is the response-surface corner at the maximum protein load of 40 g/L resin, which is the run in which a loss of resolution would show most clearly. The aggregate result of that run is therefore the result the deviation puts at risk.

The eluate pool of run 14 was retained and re-assayed for aggregate under AMV-3221, a verification-qualified size-exclusion method whose validated precision is 3.8 % relative with a limit of quantitation of 0.2 % HMW. The 3 verification injections gave a 95 % confidence interval whose half-width is 9.44 % of the mean, and the re-assayed aggregate agreed with the original result within that interval.

The response-surface model gives the same answer independently. The standardized residual of run 14 in the aggregate model is -0.48, against a largest absolute standardized residual of 1.52 across the 28 runs, so the run sits where the model predicts it and does not pull the surface. Lack of fit for that model is not significant ($p = 0.158$, Table 5.9), which would not be expected if one factorial corner had been displaced by an uncontrolled variable.

The deviation was dispositioned as retained. Run 14 was kept in the analysis and no design point was invalidated. Temperature was not a studied factor at this step, so this deviation does not establish a range for it, and the chamber remains controlled to the platform specification under SOP-2010.

14 Appendices

14.1 Appendix A — Screening design matrix and responses

Table 14.1 gives the screening design in coded units with the measured responses. The factor letters are those of Table 4.2, and the coded levels minus one, zero and plus one correspond to the low edge, the set-point and the high edge of each characterization range in Table 4.1.

Table 14.1: Screening design matrix in coded units with the measured responses. {#tbl-app-a}

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	factorial	-1	-1	-1	-1	0.5	136	0.936
2	factorial	-1	-1	-1	1	0.604	114	0.962
3	factorial	-1	-1	1	-1	0.548	135	0.94
4	factorial	-1	-1	1	1	0.818	138	0.973
5	factorial	-1	1	-1	-1	0.452	66.2	0.934
6	factorial	-1	1	-1	1	0.564	61.3	0.951
7	factorial	-1	1	1	-1	0.579	48.8	0.946
8	factorial	-1	1	1	1	0.752	67.6	0.934
9	factorial	1	-1	-1	-1	0.792	428	0.883
10	factorial	1	-1	-1	1	1.1	247	0.888
11	factorial	1	-1	1	-1	1.14	301	0.923
12	factorial	1	-1	1	1	1.79	240	0.908
13	factorial	1	1	-1	-1	0.807	105	0.895

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
14	facto-rial	1	1	-1	1	1.04	113	0.893
15	facto-rial	1	1	1	-1	1.12	115	0.884
16	facto-rial	1	1	1	1	1.83	149	0.886
17	center	0	0	0	0	0.833	153	0.929
18	center	0	0	0	0	0.829	183	0.935
19	center	0	0	0	0	0.692	129	0.91

14.2 Appendix B — Response-surface design matrix and responses

Table 14.2 gives the face-centred central composite design in coded units with the measured responses. The axial runs carry one factor at plus or minus one with the others at zero.

Table 14.2: Response-surface design matrix in coded units with the measured responses. {#tbl-app-b}

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	facto-rial	-1	-1	-1	-1	0.518	109	0.98
2	facto-rial	-1	-1	-1	1	0.593	114	0.955
3	facto-rial	-1	-1	1	-1	0.572	151	0.973
4	facto-rial	-1	-1	1	1	0.818	142	0.942
5	facto-rial	-1	1	-1	-1	0.495	59.3	0.921
6	facto-rial	-1	1	-1	1	0.616	71.9	0.946
7	facto-rial	-1	1	1	-1	0.686	60.1	0.932
8	facto-rial	-1	1	1	1	0.812	73	0.925
9	facto-rial	1	-1	-1	-1	0.768	237	0.87
10	facto-rial	1	-1	-1	1	0.973	254	0.878

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
11	factorial	1	-1	1	-1	1.15	301	0.9
12	factorial	1	-1	1	1	1.76	270	0.89
13	factorial	1	1	-1	-1	0.826	174	0.902
14	factorial	1	1	-1	1	1.12	139	0.903
15	factorial	1	1	1	-1	1.08	118	0.891
16	factorial	1	1	1	1	1.9	119	0.844
17	axial	-1	0	0	0	0.523	73.7	0.935
18	axial	1	0	0	0	1.11	186	0.879
19	axial	0	-1	0	0	0.666	225	0.92
20	axial	0	1	0	0	0.791	92.9	0.909
21	axial	0	0	-1	0	0.647	149	0.921
22	axial	0	0	1	0	0.879	148	0.914
23	axial	0	0	0	-1	0.642	162	0.935
24	axial	0	0	0	1	0.967	150	0.922
25	center	0	0	0	0	0.796	146	0.881
26	center	0	0	0	0	0.708	110	0.952
27	center	0	0	0	0	0.782	128	0.921
28	center	0	0	0	0	0.8	152	0.917

14.3 Appendix C — Full effect and coefficient tables

Table 14.3 to Table 14.5 give every term of the screening model for each response, including the terms omitted from the tables in Section 5. Table 14.6 gives the full quadratic model for step yield, which is the response not shown in Section 5.3.

Table 14.3: All screening effects on pool aggregate. {#tbl-app-c1}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*
B	-0.0185	-0.00924	0.024	-0.385	0.71	
BC	0.0152	0.00758	0.024	0.316	0.76	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
BD	-0.0131	-0.00657	0.024	-0.274	0.791	
AB	0.012	0.00602	0.024	0.251	0.808	

Table 14.4: All screening effects on pool host cell protein. {#tbl-app-c2}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	
CD	24.4	12.2	8.85	1.38	0.205	
BC	18.4	9.18	8.85	1.04	0.33	
AC	-12.5	-6.25	8.85	-0.707	0.5	
C	-9.45	-4.72	8.85	-0.534	0.608	

Table 14.5: All screening effects on step yield. {#tbl-app-c3}

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.0518	-0.0259	0.00287	-9.04	1.79e-05	***
BC	-0.0121	-0.00607	0.00287	-2.12	0.0672	
B	-0.0111	-0.00553	0.00287	-1.93	0.0899	
AD	-0.00914	-0.00457	0.00287	-1.59	0.15	
D	0.00683	0.00342	0.00287	1.19	0.268	
C	0.00666	0.00333	0.00287	1.16	0.279	
BD	-0.00566	-0.00283	0.00287	-0.988	0.352	
CD	-0.00473	-0.00236	0.00287	-0.824	0.434	
AC	0.00401	0.002	0.00287	0.699	0.504	
AB	3.38e-05	1.69e-05	0.00287	0.00589	0.995	

Table 14.6: Full quadratic model coefficients for step yield. {#tbl-app-c4}

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.917	0.00651	141	4.36e-22	***
A	-0.0307	0.00444	-6.92	1.05e-05	***
B	-0.00757	0.00444	-1.71	0.112	
C	-0.00358	0.00444	-0.807	0.434	
D	-0.00548	0.00444	-1.24	0.239	
AB	0.00803	0.00471	1.71	0.112	
AC	0.000269	0.00471	0.0571	0.955	

Term	Coef.	Std. err.	t	p-value	Sig.
AD	-0.000697	0.00471	-0.148	0.885	
BC	-0.00646	0.00471	-1.37	0.193	
BD	0.00185	0.00471	0.393	0.701	
CD	-0.00658	0.00471	-1.4	0.185	
A^2	-0.0103	0.0117	-0.882	0.394	
B^2	-0.00278	0.0117	-0.237	0.816	
C^2	0.000575	0.0117	0.049	0.962	
D^2	0.0108	0.0117	0.924	0.372	

14.4 Appendix D — Analytical methods summary

Table 14.7 lists the analyte, the technique, the method validation record and the reportable result for each measurement made in this study. Table 14.8 gives the validated performance of those methods for which a performance record is held.

Table 14.7: Analytical methods used in the cation exchange characterization study. {#tbl-app-d1}

Analyte	Technique	Method validation	Reportable result
Aggregate (high molecular weight)	Size-exclusion HPLC	AMV-3011	% HMW of total protein
Aggregate, verification assay	Size-exclusion HPLC	AMV-3221	% HMW of total protein
Host cell protein	Sandwich ELISA, polyclonal reagent	AMV-3012	ng per mg of product
Residual host cell DNA	Quantitative PCR	AMV-3014	ng per dose
Leached Protein A	Sandwich ELISA	AMV-3016	ppm of product
Protein concentration	Ultraviolet absorbance at 280 nm	AMV-3019	g/L

Table 14.8: Validated performance of the analytical methods for which a performance record is held. {#tbl-app-d2}

Method	Title	Precision (% RSD)	LOQ	LOQ unit	Accuracy (%)
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8	0.2	% HMW	99.1
AMV-3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95
AMV-3016	Leached Protein A by ELISA	6.5	0.25	ppm	97.4

Method	Title	Precision (% RSD)	LOQ	LOQ unit	Accuracy (%)
AMV-3019	Protein Concentration by A280 (UV)	1.2	0.5	g/L	99.6

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